

Cyclic Array

Time limit: 1.00 s

Memory limit: 512 MB

You are given a cyclic array consisting of n values. Each element has two neighbors; the elements at positions n and 1 are also considered neighbors.

Your task is to divide the array into subarrays so that the sum of each subarray is at most k . What is the minimum number of subarrays?

Input

The first input line contains integers n and k .

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

There is always at least one division (i.e., no value in the array is larger than k).

Output

Print one integer: the minimum number of subarrays.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq k \leq 10^{18}$

Example

Input	Output
8 5 2 2 2 1 3 1 2 1	3

Explanation: We can create three subarrays: $[2, 2, 1]$, $[3, 1]$, and $[2, 1, 2]$ (remember that the array is cyclic).